

Modular Server Events

A once-a-week world event that is never the same twice

One roll at server reset. Six days in seven, nothing happens and the day proceeds exactly as it does now. On the seventh, four modules are drawn from four separate pools, each at a randomly rolled magnitude, checked against each other for coherence, given a name derived from what was actually drawn, and published to the whole server.

This document specifies the eighty modules, the draw pipeline, the coherence gate, the naming system and the balance clamps — and hands over a verified reference implementation so none of it has to be guessed at.

80 MODULES	12,960,000 EVENT STRUCTURES	8,800 POSSIBLE NAMES	81 yrs TO A LIKELY REPEAT
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Prepared 08 September 2026 · every figure in this document is measured from a **100-year simulation** of the reference implementation, not estimated.

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1 The shape of the system

Live-service events usually fail in one of two ways. Either they are hand-authored, which means there are twelve of them and players have seen all twelve by March; or they are pure random multipliers, which means every event is "×2 something" and none of them are memorable. This system is built to avoid both.

The core idea is that an event is not *one* effect. It is a **Boon**, a **Twist**, a **Focus** and a **Flourish**, drawn independently and then reconciled. The Boon is the gift, the Twist is the complication, the Focus is where it lands, and the Flourish is the server-wide toppler. That structure is what makes a random draw read as a designed event: the grammar is fixed even though every word in the sentence is rolled.

Two consequences follow, and both are the point:

- **The combination carries the meaning, not the module.** "More gold" is dull. "More gold, but one resource family has stopped dropping entirely, and only in the top five zones, while market tax is waived" is a day players talk about — and none of those four pieces had to be written for each other.
- **The name is derived, not authored.** Nobody writes a name for an event that has never existed before. The system reads the tags off what it actually drew, works out what kind of day it is, and names it from a pool matching that kind. A player who sees *COMBAT EVENT — Marked Reaver's Hour* knows what the day is about before reading a word of the detail.

THE DESIGN CONSTRAINT THIS IS BUILT AGAINST

A player who logs in on an event day should be able to say, in one sentence, what is different — and should not be able to predict what next week's event will be. Everything below serves those two sentences.

What it costs to run

One integer roll per day, four table lookups, and a JSON blob published to a cache. There is no live tuning, no per-event content and no scheduling. Once the tables are in, the system generates events for years untouched, and the pools are additive: adding one new module to each of the four takes the structure count from 12,960,000 to over 15.7 million.

2 The daily gate

At the turn of the server day — the same boundary the Daily XP Boost already uses — the server rolls once. A value below `1/7` makes the coming day an event day. Everything else about the day is unchanged, and on a non-event day the system does nothing at all: no banner, no flag, no payload.

Two guards sit either side of the raw roll, because a memoryless 1-in-7 is not actually the player experience anyone wants:

GUARD	THRESHOLD	WHY
Drought floor	16 quiet days	A pure 1-in-7 produces a six-week gap surprisingly often — a 44-day drought appeared within ten simulated years before this guard existed. Sixteen quiet days force an event on the seventeenth. This is the only reason the observed cadence is faster than one in seven.
Streak ceiling	2 in a row	Back-to-back event days are a pleasant surprise at 1-in-49. Three in a row reads as broken, and devalues the fourth. The third consecutive roll is suppressed.

MEASURED OVER 100 SIMULATED YEARS

MEASURE	RESULT	NOTE
Events generated	5539 in 100 years	about 55 a year
Observed cadence	one every 6.59 days	faster than 7.00 because of the drought floor — this is the price of the guarantee
Events forced by the floor	421 (7.6%)	roughly one in eleven events exists because the floor fired
Longest gap between events	16 days	by construction; unbounded without the floor
Longest run of event days	2	by construction

A DECISION TO MAKE DELIBERATELY

The drought floor trades exactness for predictability: the cadence becomes 1-in-6.59 rather than 1-in-7. If the 1-in-7 figure is something you want to state publicly and have be literally true, lower the base rate to about 1-in-7.8 and let the floor bring it back — or drop the floor and accept that once a decade players will go six weeks without an event and conclude the system is broken. Our recommendation is to keep the floor and describe the cadence as "about once a week".

Why the gate gets its own sub-seed

The gate is rolled from `seed + "|gate"`, not from the same stream that draws the modules. If both came off one stream, the modules drawn would be statistically conditioned on the gate having passed — a subtle and very annoying class of bug where certain modules quietly never appear. Separate sub-seeds cost nothing and remove the question entirely.

3 The four slots

Each slot is its own pool of twenty modules. A module belongs to exactly one slot and cannot appear in another. That is a deliberate constraint rather than a limitation: it is what guarantees every event has one gift, one complication, one scope and one capstone, and it is also what prevents two modules that inflate the same number from ever landing together.

SLOT	ROLE	WHAT IT CONTRIBUTES	RULE IT ALWAYS OBEYS
A	Boon	The headline gift — the reason to log in today. Always framed as a straightforward positive.	Never carries a downside.
B	Twist	The complication that changes <i>how</i> you play rather than how much you get. Most Twists trade something away for something better.	Supplies the adjective in the event's name.
C	Focus	Where the event lands: a zone, a skill, a tier band, a four-hour window, or the whole world.	Must be able to reach both the Boon and the Twist — see §6.
D	Flourish	The server-wide capstone that applies regardless of how a player spends the day.	Always globally reachable, so nobody is locked out by build or level.

WHY SLOT SEPARATION MATTERS MORE THAN IT LOOKS

If all eighty modules sat in one pool and four were drawn from it, roughly one event in twenty would draw two modules that multiply the same quantity — and those are exactly the events that break an economy. Because Coin Flood (A1) and Silver Tongue (A13) are both Slot A, they can never co-occur. The structure does the balance work that would otherwise need a hand-maintained conflict matrix with 3,160 entries in it.

Slot D is the accessibility slot

It is worth being explicit about this, because it is easy to lose in implementation. Slots A, B and C can all end up narrow — one skill, one zone, one enemy family, four hours. If that were the whole event, a player whose evening does not intersect the Focus gets nothing at all and correctly concludes the event was not for them. Every Slot D module is globally reachable precisely so that never happens: whatever else the day is, the Flourish is for everyone.

4 The draw pipeline

Ten steps, in order. Steps 1–2 decide whether there is an event at all; 3–6 decide what it is; 7–9 make it legal and safe; 10 publishes it.

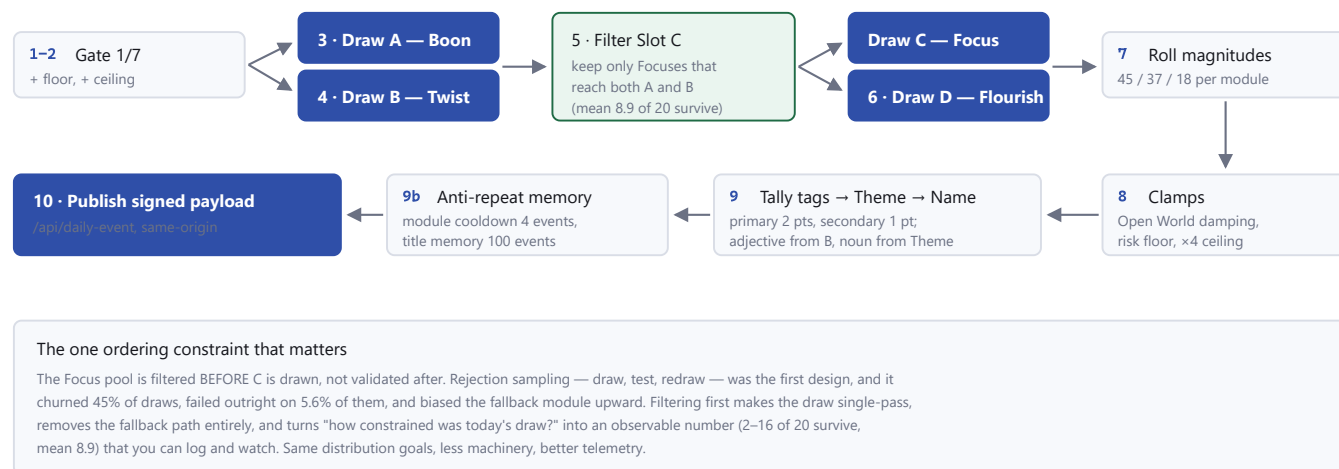


Figure 1 — the draw pipeline. Blue is a random draw; green is a filter.

Determinism

The whole pipeline is a pure function of the server-day string and the recent event history. Given the same inputs, every shard, the client, the wiki and any third-party tool compute byte-identical results. That is what makes the event verifiable rather than something players have to take on trust — and it means a desync is a visible bug rather than a silent divergence.

5 Magnitude and Rank

Each of the four drawn modules independently rolls a magnitude band, weighted **45 / 37 / 18** across I, II and III. This is the single most important multiplier on variety in the whole system: without it, there are 160,000 possible events and the first repeat is expected within about eight years of weekly events. With it, there are 12,960,000.

The bands are worth **1, 2 and 4 points**. The sum across four modules is the event's **intensity**, from 4 to 16, which maps to a public **Rank**. Rank is what players actually talk about — it is the difference between "there's an event today" and "get online, it's a Grand".

RANK	INTENSITY	THEORETICAL	MEASURED	FREQUENCY
Rank I · Faint	4–5	17.6%	24.3%	about 13.5 a year
Rank II · Rising	6–8	50.4%	46.7%	about 25.9 a year
Rank III · Greater	9–12	30.1%	27.1%	about 15.0 a year
Rank IV · Grand	13–16	2.0%	1.8%	about 1.0 a year

Measured share differs from theoretical because Open World days (\$6) damp every other module one band, which pulls a slice of the distribution downward. That is intentional: the widest events should also be the mildest.

RANK IV IS DELIBERATELY ABOUT ONCE A YEAR

A Grand event is a genuine occasion — three of four modules at their top band. At 55 events a year and a 1.8% share, players will see roughly one a year, and it should get a distinct banner, a distinct colour and an announcement. If it fires twice in eighteen months that is normal variance, not a bug; resist the urge to hand-tune it after the first cluster.

Point weighting is not linear, on purpose

Band III is worth 4 rather than 3 because a Rank should reflect the presence of an extreme, not an average. Four modules at band II (intensity 8) is a solid Rising day. Two at band III and two at band I (intensity 10) is a lopsided day with two genuinely dramatic effects, and it should — and does — rank higher.

6 The coherence gate

Independent draws produce nonsense on their own. *Rich Veins* (gathering yields) under a *Warfront* focus (combat only) is an event with no effect whatsoever — it would publish a banner, promise a bonus, and do nothing. This is the failure mode that makes procedural events feel broken, and it is not rare: without a gate, it happens on roughly a fifth of draws.

Surfaces

Every module declares which parts of the game it touches: `COMBAT`, `GATHER`, `CRAFT`, `WORKORDER`, `VILLAGE`, `TRADE`, or `GLOBAL`. `GLOBAL` intersects everything. Two modules overlap if either is `GLOBAL` or their surface sets intersect.

The rule

A Focus is legal for a given Boon and Twist if it overlaps **both** of them, and the pair is not in the hard-ban table. Rather than draw a Focus and test it, the system **filters the Slot C pool first and draws from what survives**. C12 Open World is `GLOBAL`, so the surviving pool is never empty and there is no fallback path to get wrong.

MEASURE	VALUE	READING
Legal Focuses, mean	8.9 of 20	a typical Boon/Twist pair admits about nine of the twenty Focus modules
Legal Focuses, range	2 – 16	the low end is two narrow modules pulling in different directions; the high end is two <code>GLOBAL</code> modules
Open World days	8.6%	drawn naturally from the filtered pool, not used as a fallback

The hard-ban table

Surfaces catch the great majority of incoherence structurally. A short explicit list handles what they cannot — currently one entry, and it is the extension point for anything discovered in playtest:

PAIR	REASON
B10 × C13	Two competing time gates. Night Shift makes the event dormant for the first twelve hours; Hour of Power designates a four-hour window that may fall inside the dormant half. The intersection is either incoherent or empty, and players cannot be expected to reason about it.

VERIFIED, NOT ASSUMED

The gate was tested by asserting the exclusions it should make — Warfront is correctly refused a gathering Boon, Harvest is correctly refused a combat Twist, Open World is always legal — and then by disabling each rule in turn and confirming the assertions fail. A check that has never been observed failing is not a check.

7 Naming: tags, themes, titles

Every module carries one **primary tag** and zero or more **secondary tags** from a fixed set of eleven. On an event day the system tallies them across all four drawn modules — **primary scores 2, each secondary scores 1** — and the highest total is the event's **Theme**.

Tie-breaks, in order

1. **RISK never wins a tie.** It rides along as a secondary tag on most Twists, so on a plain plurality it would take about a third of all themes and every event would be called a gamble. It is removed from any tied set — but it is allowed a *clean* plurality, which is what produces the genuinely rare gambler's day. Measured share: 4.7%, roughly two and a half a year.
2. **Slot A's primary tag wins** if it is among the leaders. The Boon is the headline, so when the draw is ambiguous the gift decides what kind of day it was.
3. **Slot D's primary tag**, then declaration order, as final deterministic backstops. These fire rarely but must exist — a tie-break that can return "no answer" is a crash on a live server at reset.

Composing the title

The name is **adjective + noun**. The adjective comes from the drawn **Twist** — four per Twist module, eighty in total — so the most distinctive mechanic of the day is what colours the name. The noun comes from the **Theme** pool, ten per theme. The format is:

```
<THEME> EVENT - <Adjective> <Noun>
<THEME> EVENT - <Adjective> <Noun> Ascendant (Rank IV) VILLAGE EVENT - Patient Bonfire Night Ascendant
<THEME> EVENT - <Adjective> <Noun> <Numeral> (a return) LOOT EVENT - Devil's Trove III
```

That is **8,800** distinct titles in the pool. A title used within the last 100 events is redrawn; beyond that window a repeat is permitted and **numbered from the archive**. This turns the one genuinely unavoidable collision into a feature — players remember the third Ravening Blood Moon, and a returning name reads as a sequel rather than a bug. Over 100 simulated years and 5539 events, **0 unnumbered duplicate names** occurred.

Theme frequencies and noun pools

THEME	SHARE	FREQUENCY	NOUN POOL (10 EACH)
COMBAT	21.7%	12.0 a year	Blood Moon · Warcry · Red Tide · Iron Harvest · Siegefall · Bonefield · Long Battle · Hollow Crown · Ashen Vigil · Reaver's Hour
GATHERING	6.4%	3.5 a year	Green Tide · First Harvest · Deep Seam · Rootfall · Amber Season · Long Gather · Wildgrowth · Stonebloom · Sapmoon · Full Cart
CRAFTING	7.7%	4.3 a year	Forgefire · Anvil Chorus · Emberwork · Quenching · Maker's Hour · Ashfall Forge · Loomsong · Bellowsnight · Fine Seam · Hammerfall
ECONOMY	15.6%	8.6 a year	Gilded Hour · Grand Exchange · Coinfall · Merchant's Moon · Ledger · Golden Wake · Copper Rain · Long Market · Vault Season · Purse-Cut
WORKORDER	5.6%	3.1 a year	Muster Roll · Quota · Standing Order · Requisition · Warrant Day · Foreman's Call · Backlog · Docket · Signed and Sealed · Order of the Day

THEME	SHARE	FREQUENCY	NOUN POOL (10 EACH)
LOOT	12.7%	7.1 a year	Spoils · Glittering Hour · Hoardfall · Treasurewake · Bright Take · Cache Season · Windfall · Full Pack · Trove · Scattered Riches
XP	10.9%	6.1 a year	Enlightenment · Long Lesson · Ascension · Mindfire · Steep Climb · Revelation · Whetstone · Quickening · Insight · Breakthrough
EXPLORATION	3.3%	1.8 a year	Wayfare · Open Road · Farwander · Uncharted · Horizonfall · Long Path · Trailblaze · Strange Country · Far Marches · Driftlands
VILLAGE	4.2%	2.3 a year	Raising · Hearthlight · Common Good · Foundation Day · Rooftree · Village Wake · Cornerstone · Bonfire Night · Long Table · Homecoming
SOCIAL	7.2%	4.0 a year	Convergence · Gathering · Assembly · Chorus · Great Meet · Commonwake · Many Hands · Shared Hour · Congress · Fellowship
RISK	4.7%	2.6 a year	Gamble · Fool's Wager · Knife-Edge · Long Odds · Tempest · Coin-Toss · Brinkfall · Wild Hunt · Precipice · Snake Eyes

The distribution is uneven by design and it reflects the game: combat is the largest surface in IdleWorlds, so COMBAT themes most often. What matters is that no theme is starved — the rarest, EXPLORATION, still lands about twice a year, which is often enough to feel like part of the rotation and rare enough to feel like an occasion.

8 Anti-repeat memory

Raw combination count is not the same as felt variety. An event whose modules all appeared last week feels repetitive even if the exact four-tuple is new. Three cheap memories fix that; all three are small enough to keep in the same row as the event archive.

MEMORY	WINDOW	EFFECT
Module cooldown	4 events	A module that has appeared in any of the last four events is removed from its pool. At about 55 events a year that is roughly a month's rest, and it is what stops the same Boon showing up three times in five weeks.
Title memory	100 events	A title used in the last 100 events is redrawn, up to eight attempts.
Signature memory	archive	The full <code>A2-B10-C4-D1</code> -style signature is stored per event. Used for the numeral suffix, for "this has happened before" in the archive UI, and as the source for D20 Echo of Days.

THE COOLDOWN SLIGHTLY RAISES COLLISION ODDS — KNOWINGLY

Blocking four modules per slot shrinks each pool from 20 to as few as 16, which raises the chance of an exact structural repeat by roughly 1.9×. Over 100 simulated years the system produced **8 exact structural repeats** across 5539 events, against about 1.2 expected with no cooldown. That is the right trade: an exact repeat a century apart is invisible, while the same module twice in a fortnight is not.

9 How many events are there really?

The brief asked for enough combinations that no two events are ever likely to be the same. Here is the honest accounting, with the weakest claim first.

COUNTING	COUNT	WHAT IT INCLUDES
Module identity only	160,000	20^4 . Not sufficient on its own — at 55 events a year, a repeat becomes more likely than not inside nine years.
Structure — modules + ranks	12,960,000	$20^4 \times 3^4$. This is the number to quote: it is what a player would perceive as "the same event".
With rolled targets	56,800,501,560	Includes which zone, which skill, which enemy family, which window. Genuine variety, but dominated by C14's ordered four-skill rotation; we do not lead with this number.
Distinct names	8,800	80 Twist adjectives $\times$ 110 Theme nouns, before the archive numeral.

When would a repeat actually happen?

Against the structural count of 12,960,000, and by the birthday bound, the expected number of events before a first exact structural repeat is about **4,512** — roughly **81 years** of weekly events. Over a far more realistic ten-year horizon, the probability of *any* exact structural repeat occurring at all is **1.18%**.

The 100-year simulation bears this out: 8 structural repeats in 5539 events, and even those differ in their rolled targets. In practice, no player will ever see the same event twice.

WHERE THE VARIETY ACTUALLY COMES FROM

Note which factor is load-bearing. Going from 20 modules per slot to 25 would multiply the structure count by 2.4 $\times$. Adding the magnitude roll multiplied it by **81 $\times$** . If schedule pressure forces a cut, cut modules — never cut magnitude bands.

10 Balance clamps

Slot separation prevents most stacking, but not all of it. Four clamps run after the draw and before publication.

CLAMP	RULE
Open World damping	If C12 Open World is drawn, every other module drops one magnitude band (never below I). Removing all restriction has to cost something, or Open World becomes the strictly best Focus. C12's own band is pinned to I before intensity is summed — its three bands are identical text, so letting its roll into the sum scored the mildest days of the year as some of the strongest. Measured effect: Open World days average intensity 4.6 against 7.6 elsewhere, which is correct.
Risk floor	A10 Second Wind at band III removes the death penalty entirely. Beside B11 Glass Cannon — the Twist whose whole point is that you might die — that neutralises the day's only real danger. When both are drawn, A10 is held to band I. This is the template for any future conditional clamp: reduce a magnitude, do not ban a pair.
Ceiling	No single quantity may exceed $\times 4$ from all event sources combined ($\times 6$ at Rank IV). Applied at the point of effect, after every multiplier is resolved. This is the backstop that means a new module can be added without re-deriving every interaction it might have.
Kill switch	A single server flag suppresses the day's event and reverts to normal rules without a deploy. Non-negotiable for a system that generates content nobody reviewed in advance.

THE CLAMP THAT WILL BE NEEDED AND IS NOT HERE YET

Nothing in this spec models the interaction between event multipliers and existing permanent player bonuses — gear Gold Find, Item Find, add-on buffs, set bonuses. A player at the top of the gear curve will get materially more out of Coin Flood than a new one, which is fine, and out of Fortune's Eye specifically, which may not be. Decide before launch whether event bonuses are additive with gear stats or multiplicative; the $\times 4$ ceiling assumes additive.

11 Module table — Slot A

Slot A — the Boon

The headline gift. What the announcement leads with, and the reason a player logs in today rather than tomorrow. Slot A is always a straightforward positive; it never carries a downside.

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
A1	Coin Flood Every gold source on the server pays more.	ECONOMY · LOOT reaches: GLOBAL	I +25% gold II +60% gold III +120% gold	—
A2	Rich Veins Gathering nodes give up a second helping.	GATHERING · ECONOMY reaches: GATHER	I 10% double-yield II 20% double-yield III 35% double-yield	—
A3	Master's Hand Crafts occasionally come off the bench twice.	CRAFTING · ECONOMY reaches: CRAFT	I 8% free extra craft II 15% free extra craft III 25% free extra craft	—
A4	Battle Fervour Combat experience surges.	COMBAT · XP reaches: COMBAT	I +30% combat XP II +70% combat XP III +150% combat XP	—
A5	Deep Pockets Item Find rises for everyone, equipped or not.	LOOT · COMBAT reaches: COMBAT, GATHER	I +15 Item Find II +35 Item Find III +75 Item Find	—
A6	Fortune's Eye Gold Find rises for everyone, equipped or not.	LOOT · ECONOMY reaches: COMBAT, GATHER	I +20 Gold Find II +45 Gold Find III +90 Gold Find	—
A7	Quartermaster's Ledger Work Order payouts are inflated.	WORKORDER · ECONOMY reaches: WORKORDER	I +30% WO reward II +65% WO reward III +130% WO reward	—
A8	Tireless Every non-combat action ticks faster.	GATHERING · CRAFTING reaches: CRAFT, GATHER	I +10% action speed II +20% action speed III +35% action speed	—
A9	Warband Warfare climbs across the roster.	COMBAT reaches: COMBAT	I +10% Warfare II +22% Warfare III +45% Warfare	—
A10	Second Wind Recovery is fast and dying costs less.	COMBAT · RISK reaches: COMBAT	I ×2 recovery, –50% death penalty II ×3 recovery, –75% death penalty III ×5 recovery, no death penalty	—
A11	Salvager's Luck Enhancement materials fall out of ordinary actions.	LOOT · CRAFTING reaches: COMBAT, GATHER	I 5% bonus material II 10% bonus material III 18% bonus material	—

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
A12	Foreman's Whistle Building Parts come quicker and cheaper.	VILLAGE · CRAFTING reaches: VILLAGE	I +25% Parts XP, –10% bill II +55% Parts XP, –20% bill III +110% Parts XP, –35% bill	—
A13	Silver Tongue Vendors buy high and sell low.	ECONOMY reaches: GLOBAL, TRADE	I –10% buy / +15% sell II –20% buy / +30% sell III –33% buy / +60% sell	—
A14	Apprentice's Boon Your three lowest skills catch up fast.	XP · SOCIAL reaches: GLOBAL	I +50% XP on lowest 3 II +110% XP on lowest 3 III +220% XP on lowest 3	—
A15	Restless Traveller Zone gates relax and travel is free.	EXPLORATION · COMBAT reaches: COMBAT, GLOBAL	I –3 zone level req II –6 zone level req III –10 zone level req	—
A16	Overflow The idle bucket holds far more before it spills.	ECONOMY · XP reaches: GLOBAL	I +4h idle cap II +8h idle cap III +16h idle cap	—
A17	Gem-Sight Gems and sockets surface everywhere.	LOOT · CRAFTING reaches: COMBAT, GATHER	I ×2 gem drops II ×3 gem drops III ×5 gem drops	—
A18	Alchemist's Surplus Potions last longer and are often not spent.	CRAFTING · COMBAT reaches: COMBAT, CRAFT	I +50% duration, 15% not consumed II +100% duration, 30% not consumed III +200% duration, 50% not consumed	—
A19	Village Rally Every add-on and building buff is amplified.	VILLAGE · SOCIAL reaches: GLOBAL, VILLAGE	I +25% add-on buffs II +50% add-on buffs III +100% add-on buffs	—
A20	Task Bounty Task-tier completions pay a bounty.	WORKORDER · XP reaches: WORKORDER	I ×1.5 task rewards II ×2 task rewards III ×3 task rewards	—

12 Module table — Slot B

Slot B — the Twist

The complication that makes the day feel unlike any other. Twists change how you play, not just how much you get. This is the slot that supplies the adjective in the event name.

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
B1	Swarm Encounters bring friends. Per-kill rewards are unchanged, so the day is strictly denser, not strictly richer.	COMBAT · RISK reaches: COMBAT	I +1 enemy per fight II +2 enemies per fight III +4 enemies per fight	—
B2	Elite Vanguard A share of spawns are Elites: double health, two-and-a-half times the take.	COMBAT · LOOT reaches: COMBAT	I 10% Elites II 20% Elites III 35% Elites	—
B3	Brittle Steel Enhancement is far riskier, but every success lands one level deeper.	CRAFTING · RISK reaches: CRAFT	I +15pp failure, +1 level on success II +25pp failure, +1 level on success III +40pp failure, +1 level on success	—
B4	Blind Draw Every drop rolls twice and keeps the better roll. Nothing is previewed anywhere in the UI until it is claimed.	LOOT · RISK reaches: GLOBAL	I best of 2 on rares II best of 2 on all drops III best of 3 on all drops	—
B5	Iron Rations Healing is halved; the survivors learn faster for it.	COMBAT · RISK reaches: COMBAT	I –50% healing, +40% combat XP II –50% healing, +80% combat XP III –50% healing, +150% combat XP	—
B6	Rush Orders Shorter fuses, bigger quotas.	WORKORDER · RISK reaches: WORKORDER	I –25% timer, +50% quantity II –40% timer, +90% quantity III –60% timer, +150% quantity	—
B7	Heavy Haul Carry less, gather more. Bank trips become the real cost.	GATHERING · RISK reaches: GATHER	I –30% carry, +40% yield II –45% carry, +80% yield III –60% carry, +150% yield	—
B8	Contested Ground A server-wide kill counter runs in the focused area; everyone shares the escalating bonus it pays out.	SOCIAL · COMBAT reaches: COMBAT	I +5% per 10k kills, cap +25% II +5% per 10k kills, cap +50% III +5% per 5k kills, cap +75%	—
B9	Tempered Tools Gathering is slower per action and much larger per action, which rewards long uninterrupted sessions over check-ins.	GATHERING · RISK reaches: GATHER	I –25% speed, ×2 yield II –35% speed, ×2.5 yield III –50% speed, ×3.5 yield	—

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
B10	Night Shift Nothing happens for the first half of the server day. Then everything happens harder.	RISK · SOCIAL reaches: GLOBAL	I dormant 12h, then ×1.5 II dormant 12h, then ×2 III dormant 12h, then ×2.5	—
B11	Glass Cannon Damage dealt and damage taken both climb, dealt slightly faster.	COMBAT · RISK reaches: COMBAT	I +50% dealt / +40% taken II +100% dealt / +80% taken III +200% dealt / +150% taken	—
B12	Tithe A slice of every coin earned goes into a server pot, paid back at reset to contributors at one and a half times what they put in.	ECONOMY · SOCIAL reaches: GLOBAL	I 10% tithe II 15% tithe III 25% tithe	—
B13	Rivalry A public leaderboard opens for the themed activity and pays the top of it at reset.	SOCIAL · XP reaches: GLOBAL	I top 25 rewarded II top 100 rewarded III top 500 rewarded	—
B14	Unstable Enchantment Enchantments on equipped gear reroll to a random enchantment of the same tier on a timer. Nothing is destroyed; the loadout simply will not sit still.	CRAFTING · RISK reaches: GLOBAL	I reroll every 4h II reroll every 2h III reroll every 1h	—
B15	Wandering Merchant A merchant with a randomly drawn stock sets up in one zone and moves every few hours.	ECONOMY · EXPLORATION reaches: GLOBAL, TRADE	I 6 lines at 60% price II 10 lines at 50% price III 15 lines at 40% price	zone (34)
B16	Scarcity One resource family stops dropping entirely. Anything already made from it is worth a fortune.	ECONOMY · GATHERING reaches: CRAFT, GATHER, TRADE	I ×2 value on affected goods II ×3 value on affected goods III ×5 value on affected goods	resource family (8)
B17	Conscription Finishing a Work Order reassigns you to a different skill. Each link in the chain pays better than the last.	WORKORDER · RISK reaches: WORKORDER	I +20% per chain link II +35% per chain link III +50% per chain link	—
B18	Momentum Repeating the same action stacks a bonus. Switching resets it to zero.	GATHERING · CRAFTING reaches: CRAFT, GATHER	I +1% per action, cap +25% II +1% per action, cap +50% III +1% per action, cap +100%	—
B19	Hunter's Mark One enemy family is marked. It is worth several times normal; everything else is worth rather less.	COMBAT · LOOT reaches: COMBAT	I ×3 marked / ×0.6 rest II ×4 marked / ×0.6 rest III ×6 marked / ×0.6 rest	enemy family (8)
B20	Doubling Down Any reward can be pushed: better than even odds to double it, otherwise it is gone. Opt-in, per reward.	RISK · LOOT reaches: GLOBAL	I 55% to double, 3 pushes/h II 60% to double, 5 pushes/h III 65% to double, unlimited	—

13 Module table — Slot C

Slot C — the Focus

Where the event lands: a zone, a skill, a tier band, a four-hour window, the whole world. The Focus is what stops every event being "everything, everywhere, more" and is the slot the coherence gate polices.

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
C1	Zone Lock One zone becomes the epicentre; the rest of the world is damped.	EXPLORATION reaches: COMBAT, GATHER	I ×1.5 inside / ×0.75 outside II ×2 inside / ×0.75 outside III ×3 inside / ×0.75 outside	zone (34)
C2	Tier Band Only a five-tier band of content is affected.	EXPLORATION · COMBAT reaches: COMBAT, GATHER	I ×1.5 in band II ×2 in band III ×2.5 in band	tier band (30)
C3	Skill Trio Three skills carry the event; the other seven have a normal day.	XP · GATHERING reaches: CRAFT, GATHER, VILLAGE	I ×1.5 on the trio II ×2 on the trio III ×2.5 on the trio	three skills (120)
C4	Spotlight One skill only, and it is turned up hard.	XP reaches: CRAFT, GATHER, VILLAGE	I ×2 on that skill II ×2.5 on that skill III ×3 on that skill	skill (10)
C5	Element Ascendant One damage type is ascendant; event combat effects scale with using it.	COMBAT reaches: COMBAT	I ×1.5 with that type II ×2 with that type III ×2.5 with that type	damage type (8)
C6	Slotwork One equipment slot has its stats multiplied for the day.	COMBAT · CRAFTING reaches: COMBAT	I ×1.5 slot stats II ×2 slot stats III ×2.5 slot stats	gear slot (12)
C7	Set Surge Set bonuses count extra pieces you are not wearing.	COMBAT · CRAFTING reaches: COMBAT	I +1 phantom piece II +2 phantom pieces III all sets active at 4 pieces	—
C8	Hearthbound Effects run through the village and scale with what has been built.	VILLAGE · SOCIAL reaches: VILLAGE	I +4% per add-on, cap +40% II +6% per add-on, cap +60% III +8% per add-on, cap +80%	—
C9	Estate Magnitude scales off housing tier, so the event rewards long-term investment.	VILLAGE · ECONOMY reaches: GLOBAL, VILLAGE	I +10% per housing tier II +15% per housing tier III +20% per housing tier	—
C10	Sanctuary Only content at or below your level minus ten is affected. A deliberate catch-up lens.	XP · SOCIAL reaches: GLOBAL	I ×2 on backfill content II ×3 on backfill content III ×4 on backfill content	—
C11	Deep End The top five zones only.	COMBAT · EXPLORATION reaches: COMBAT	I ×2 at the top II ×2.5 at the top III ×3 at the top	—

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
C12	Open World No restriction at all — but every other module drops one magnitude band (never below I). The safety valve and the wide, gentle day.	— neutral — reaches: GLOBAL	I unrestricted, -1 band II unrestricted, -1 band III unrestricted, -1 band	—
C13	Hour of Power A four-hour window is announced at reset. Inside it the event triples; outside it the event is halved.	RISK · SOCIAL reaches: GLOBAL	I ×3 in window / ×0.5 out II ×4 in window / ×0.5 out III ×5 in window / ×0.5 out	4h window (20)
C14	Rotation The focus walks through four drawn targets, six hours each.	EXPLORATION · XP reaches: GLOBAL	I ×1.5 on the active target II ×2 on the active target III ×2.5 on the active target	four targets, ordered (5040)
C15	Warfront Combat only.	COMBAT reaches: COMBAT	I ×1.5 in combat II ×2 in combat III ×2.5 in combat	—
C16	Harvest Gathering only.	GATHERING reaches: GATHER	I ×1.5 gathering II ×2 gathering III ×2.5 gathering	—
C17	Forge Crafting and refining only.	CRAFTING reaches: CRAFT	I ×1.5 crafting II ×2 crafting III ×2.5 crafting	—
C18	Ledger Work Orders and task tiers only.	WORKORDER · ECONOMY reaches: WORKORDER	I ×1.5 on orders II ×2 on orders III ×2.5 on orders	—
C19	Muster Effects scale with how many players are in your zone right now.	SOCIAL · COMBAT reaches: COMBAT	I +2% each, cap +40% II +3% each, cap +50% III +5% each, cap +60%	—
C20	Watermark Effects scale with how far past your own previous-day total you get. Competes with nobody but yesterday.	SOCIAL · XP reaches: GLOBAL	I +1% per 5% over, cap +50% II +1% per 4% over, cap +75% III +1% per 3% over, cap +100%	—

14 Module table — Slot D

Slot D — the Flourish

The server-wide capstone that applies no matter how a player spends the day. Every Flourish is globally reachable by design, so no player is locked out of an event by their build or their level.

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
D1	Great Learning All experience, every skill, all day.	XP reaches: GLOBAL	I ×1.5 all XP II ×2 all XP III ×2.5 all XP	—
D2	Spoils Drop quantities across the board.	LOOT reaches: GLOBAL	I ×1.5 drop quantity II ×2 drop quantity III ×3 drop quantity	—
D3	Wider Boon Today's Daily XP Boost covers more than its usual three skills.	XP reaches: GLOBAL	I 5 skills boosted II 6 skills boosted III all 10 skills boosted	—
D4	Free Rerolls Reroll tasks and Work Orders at no cost.	WORKORDER reaches: GLOBAL	I 3 rerolls II 5 rerolls III unlimited rerolls	—
D5	Event Currency A token drops server-wide and an event shop stays open for 48 hours past reset, so nobody is punished for sleeping through the day.	ECONOMY · SOCIAL reaches: GLOBAL	I 6 shop lines II 10 shop lines III 15 shop lines	—
D6	Regalia A title and banner named for this exact event, awarded on participation and never reissued. The permanent record that this day happened.	SOCIAL reaches: GLOBAL	I title II title + banner III title + banner + frame	—
D7	Server Goal One global bar the whole server fills together.	SOCIAL reaches: GLOBAL	I 1 reward tier II 2 reward tiers III 3 reward tiers	—
D8	Milestone Chain Personal milestones through the day, each paying more than the last.	XP · LOOT reaches: GLOBAL	I 3 milestones II 5 milestones III 7 milestones	—
D9	Grace Guaranteed enhancement successes, banked per account.	CRAFTING reaches: GLOBAL	I 1 guaranteed success II 2 guaranteed successes III 3 guaranteed successes	—
D10	Gem Rain Gems fall from everything.	LOOT · CRAFTING reaches: GLOBAL	I ×2 gems II ×3 gems III ×5 gems	—
D11	Amnesty Every gold sink is switched off: travel, repair, respec, listing fees.	ECONOMY reaches: GLOBAL	I travel + repair free II travel + repair + respec free III all sinks free	—

ID	MODULE	TAGS	MAGNITUDE BANDS	ROLLS A TARGET
D12	Second Ledger Extra concurrent Work Order slots.	WORKORDER reaches: GLOBAL	I +1 slot II +2 slots III +3 slots	—
D13	Construction Blitz The village goes up fast and cheap.	VILLAGE reaches: GLOBAL	I ×2 speed, −10% Parts II ×3 speed, −20% Parts III ×5 speed, −30% Parts	—
D14	The Rift A temporary zone opens, level-scaled to each player, with its own loot table that closes at reset.	EXPLORATION · COMBAT reaches: GLOBAL	I rift, 1 floor II rift, 3 floors III rift, 5 floors	—
D15	Loot Echo Drops can duplicate, and the duplicate can duplicate.	LOOT reaches: GLOBAL	I 8% echo II 12% echo III 20% echo	—
D16	First Blood The first kills of each hour are guaranteed rares — a shape that rewards checking in rather than parking.	LOOT · COMBAT reaches: GLOBAL	I first 5 per hour II first 10 per hour III first 20 per hour	—
D17	Overreach Your highest skill counts as several levels higher for requirements and yields.	XP reaches: GLOBAL	I +3 effective levels II +5 effective levels III +8 effective levels	—
D18	Overtime Idle and offline accrual pays at the full active rate for the event day.	ECONOMY reaches: GLOBAL	I 100% offline rate, 8h II 100% offline rate, 16h III 100% offline rate, 24h	—
D19	Trader's Holiday Market tax is waived and vendors pay a premium.	ECONOMY reaches: GLOBAL	I no tax, +20% sell II no tax, +40% sell III no tax, +75% sell	—
D20	Echo of Days One module from a randomly chosen past event returns as a fifth effect at magnitude I. The archive feeds itself.	RISK · SOCIAL reaches: GLOBAL	I 1 revived module at I II 1 revived module at I III 2 revived modules at I	a past module (80)

15 Name pools

Twist adjectives — four per Slot B module

The adjective is drawn from whichever Twist was rolled, which is what ties the name to the day's most distinctive mechanic.

ID	TWIST	ADJECTIVE POOL
B1	Swarm	Teeming · Endless · Ravening · Boundless
B2	Elite Vanguard	Crowned · Vanguard's · Imperial · Warlord's
B3	Brittle Steel	Brittle · Tempered · Reckless · Hammered
B4	Blind Draw	Veiled · Hidden · Shrouded · Unseen
B5	Iron Rations	Starving · Lean · Gaunt · Hollow
B6	Rush Orders	Hurried · Frantic · Breakneck · Clamouring
B7	Heavy Haul	Laden · Burdened · Groaning · Overladen
B8	Contested Ground	Contested · Warring · Thronged · Embattled
B9	Tempered Tools	Patient · Slow-Burning · Deliberate · Enduring
B10	Night Shift	Nocturnal · Midnight · Twilit · Waking
B11	Glass Cannon	Brittle-Boned · Reckless · Sundering · Frenzied
B12	Tithe	Tithed · Sworn · Covenant · Pledged
B13	Rivalry	Rival · Proving · Contested · Crowning
B14	Unstable Enchantment	Unstable · Shifting · Wild · Unbound
B15	Wandering Merchant	Wandering · Vagrant · Gilded · Far-Roaming
B16	Scarcity	Barren · Withered · Hungering · Fallow
B17	Conscription	Conscript · Levied · Marching · Summoned
B18	Momentum	Relentless · Unbroken · Ceaseless · Driving
B19	Hunter's Mark	Hunted · Marked · Stalking · Quarry's
B20	Doubling Down	Gambler's · Fated · Chancer's · Devil's

Theme noun pools are listed in the table in §7.

16 Five worked examples

These are not illustrations written by hand. They are five real consecutive-calendar draws taken from the reference implementation seeded with actual server-day strings, selected from a two-year run to show a spread of theme, rank and mechanic. Every figure — the tag tally, the Focus pool size, the rolled targets, the engine log — is the engine's own output.

EXAMPLE 1 · SERVER DAY 2026-09-14 · SEED 2026-09-14

XP EVENT — Shrouded Steep Climb

Rank II · Rising · intensity 7 · Focus pool 20 of 20

SLOT	ID	MODULE AND ROLLED EFFECT	BAND	TAGS	PTS
A	A14	Apprentice's Boon — +50% XP on lowest 3	I	XP · SOCIAL	1
B	B4	Blind Draw — best of 2 on rares	I	LOOT · RISK	1
C	C10	Sanctuary — ×4 on backfill content	III	XP · SOCIAL	4
D	D7	Server Goal — 1 reward tier	I	SOCIAL	1

TAG TALLY

SOCIAL 4

XP 4

LOOT 2

RISK 1

THEME

XP — 2-way tie at 4 points, broken by Slot A's primary tag.

NAME

adjective **Shrouded** from the B4 pool, noun **Steep Climb** from the XP pool.

ENGINE LOG

- Focus pool: 20 of 20 Slot C modules reach both A14 and B4.

Reads as — a quiet, generous catch-up day. Your three weakest skills gain half again as fast, and anything at least ten levels below you pays four times — so the day is aimed squarely at the alt, the neglected skill and the returning player. Nothing is previewed before you claim it, and the whole server is filling one shared bar in the background. Low intensity, high goodwill.

VILLAGE EVENT — Patient Bonfire Night Ascendant

Rank IV · Grand · intensity 13 · Focus pool 6 of 20

SLOT	ID	MODULE AND ROLLED EFFECT	BAND	TAGS	PTS
A	A12	Foreman's Whistle — +110% Parts XP, −35% bill	III	VILLAGE · CRAFTING	4
B	B9	Tempered Tools — −50% speed, ×3.5 yield	III	GATHERING · RISK	4
C	C4	Spotlight — ×2 on that skill rolled target: Tailoring	I	XP	1
D	D18	Overtime — 100% offline rate, 24h	III	ECONOMY	4

TAG TALLY ECONOMY 2 GATHERING 2 **VILLAGE 2** XP 2 CRAFTING 1 RISK 1

THEME **VILLAGE** — 4-way tie at 2 points, broken by Slot A's primary tag.

NAME adjective **Patient** from the B9 pool, noun **Bonfire Night** from the VILLAGE pool, with **Ascendant** appended for Rank IV.

ENGINE LOG

- Focus pool: 6 of 20 Slot C modules reach both A12 and B9.

Reads as — the rare Grand day, and it is a builder's one. Parts XP more than doubles while the assembly bill drops a third; gathering is half speed but three and a half times the haul, which turns Tailoring — the rolled spotlight, doubled again — into the only sane thing to do for twenty-four hours. Offline accrual pays the full active rate all day, so this is the one event a player can genuinely sleep through and still win. Note the four-way tag tie broken by Slot A's primary.

COMBAT EVENT — Marked Reaver's Hour

Rank II · Rising · intensity 6 · Focus pool 11 of 20

SLOT	ID	MODULE AND ROLLED EFFECT	BAND	TAGS	PTS
A	A4	Battle Fervour — +70% combat XP	II	COMBAT · XP	2
B	B19	Hunter's Mark — ×3 marked / ×0.6 rest rolled target: Aberrant	I	COMBAT · LOOT	1
C	C7	Set Surge — +1 phantom piece	I	COMBAT · CRAFTING	1
D	D15	Loot Echo — 12% echo	II	LOOT	2

TAG TALLY **COMBAT 6** LOOT 3 CRAFTING 1 XP 1

THEME **COMBAT** — clear plurality at 6 points.

NAME adjective **Marked** from the B19 pool, noun **Reaver's Hour** from the COMBAT pool.

ENGINE LOG

- Focus pool: 11 of 20 Slot C modules reach both A4 and B19.

Reads as — the shape the brief asked for, almost exactly. Three of four modules are tagged COMBAT, so the name comes from the combat pool. Aberrants are worth triple and everything else is worth rather less, which pushes the whole server toward the same handful of zones; set bonuses count a phantom piece, so half-finished sets suddenly come online for a day; and one drop in eight quietly duplicates itself.

WORKORDER EVENT — Marching Standing Order

Rank I · Faint · intensity 5 · Focus pool 5 of 20

SLOT	ID	MODULE AND ROLLED EFFECT	BAND	TAGS	PTS
A	A20	Task Bounty — ×2 task rewards	II	WORKORDER · XP	2
B	B17	Conscription — +20% per chain link	I	WORKORDER · RISK	1
C	C12	Open World — unrestricted, -1 band	I	neutral	1
D	D13	Construction Blitz — ×2 speed, -10% Parts	I	VILLAGE	1

TAG TALLY **WORKORDER 4** VILLAGE 2 RISK 1 XP 1

THEME **WORKORDER** — clear plurality at 4 points.

NAME adjective **Marching** from the B17 pool, noun **Standing Order** from the WORKORDER pool.

ENGINE LOG

- Focus pool: 5 of 20 Slot C modules reach both A20 and B17.
- C12 Open World: own band pinned to I; every other module drops one band.

Reads as — the safety valve doing its job. Task Bounty and Conscription are both Work Order modules, so the Focus pool narrowed to five — and the draw came up Open World, which lifts every restriction and damps every other module one band in exchange. The result is a mild, wide day that touches everyone: double task rewards, Work Orders that keep reassigning you to new skills for a compounding bonus, and the village going up at double speed underneath it.

RISK EVENT — Midnight Brinkfall

Rank III · Greater · intensity 10 · Focus pool 9 of 20

SLOT	ID	MODULE AND ROLLED EFFECT	BAND	TAGS	PTS
A	A2	Rich Veins — 10% double-yield	I	GATHERING · ECONOMY	1
B	B10	Night Shift — dormant 12h, then ×2.5	III	RISK · SOCIAL	4
C	C4	Spotlight — ×3 on that skill rolled target: Foraging	III	XP	4
D	D20	Echo of Days — 1 revived module at I	I	RISK · SOCIAL	1

TAG TALLY **RISK 4** GATHERING 2 SOCIAL 2 XP 2 ECONOMY 1

THEME **RISK** — clear plurality at 4 points.

NAME adjective **Midnight** from the B10 pool, noun **Brinkfall** from the RISK pool.

ENGINE LOG

- Focus pool: 9 of 20 Slot C modules reach both A2 and B10.

Reads as — the gambler’s day, and the one that shows the archive feeding itself. Nothing at all happens for the first twelve hours — then the event wakes at ×2.5 for the back half, with Foraging tripled inside it. Echo of Days drags one module out of a past event and bolts it on as a fifth effect. RISK only takes a theme on a clean plurality, which is why this reads as a RISK event rather than a gathering one.

17 Reference implementation

The full engine accompanies this document as a single Python file with no dependencies: the eighty modules, the pipeline, the naming system, the clamps and the simulation harness that produced every number quoted here. It is intended as an executable specification — the tables in §11–§14 are generated from it, so they cannot drift.

The PRNG

Determinism across server, client and any third-party tool requires everyone to agree on the random stream, which means specifying the generator rather than using each platform's built-in. The system uses **FNV-1a** to hash the seed string to 32 bits and **mulberry32** to expand it.

```
function fnv1a(s){ let h = 0x811C9DC5;
  for (const c of Buffer.from(s, 'utf8')) { h ^= c; h = Math.imul(h, 0x01000193) >>> 0; }
  return h >>> 0; }

function mulberry32(a){ return function(){
  a = (a + 0x6D2B79F5) >>> 0;
  var t = a;
  t = Math.imul(t ^ t >>> 15, 1 | t);
  t = t + Math.imul(t ^ t >>> 7, 61 | t) ^ t;
  return ((t ^ t >>> 14) >>> 0) / 4294967296; }; }

// seeds
const gate  = mulberry32(fnv1a(serverDay + '|gate')); // 1-in-7 check
const draw  = mulberry32(fnv1a(serverDay));           // module draws
```

THE PORT IS VERIFIED, NOT ASSUMED

The Python reference and this JavaScript were run side by side over four seed strings — including a real server-day string and a gate sub-seed — and compared to twelve decimal places across six successive draws each. All 24 values match exactly. The subtlety worth knowing about: `t + Math.imul(...)` in JS is a float64 addition of two int32 values that is only truncated to 32 bits by the following `^`. A port that masks to 32 bits too early diverges after a handful of draws, and the divergence is invisible until two clients disagree about what day it is.

Draw order is part of the contract

Because every draw consumes from one stream, the **order of calls is load-bearing**. A, then B, then the filtered C, then D; magnitude immediately after each module; rolled target immediately after magnitude. Reordering any of these produces a different but equally valid event, which means a refactor that looks purely cosmetic will silently change every future event and break agreement with any client that was not redeployed at the same instant. Freeze the order, and put a comment saying so next to it.

18 The published event contract

The event must be published as machine-readable data at a same-origin endpoint. This is the single most important integration requirement in this document, and it is worth being blunt about why.

THE LESSON ALREADY PAID FOR

The Daily XP Boost has no offline answer. Because it is not published anywhere machine-readable, external tools have to scrape a wiki page through third-party CORS readers, and when that fails the only correct behaviour is to display "unavailable" — because a wrong guess applies a bonus to skills that do not have one and withholds it from ones that do, and every XP-per-hour figure downstream inherits the error while still looking plausible.

A modular event system is **strictly worse** in this respect: there are four simultaneous effects instead of one, several are multiplicative, and no external tool can possibly infer them. If the event is not published as data, every calculator in the ecosystem is silently wrong for one day in seven. Publishing it is a few lines of JSON and it prevents that entirely.

REQUIREMENTS

- **Same-origin**, e.g. `/api/daily-event`. An absolute URL serving 200 with no `Access-Control-Allow-Origin` header is discarded by every browser and looks identical to "no event today".
- **Numbers, not prose**. `"goldMultiplier": 1.6`, not `"Gold is increased by 60%"`. Prose has to be parsed, and parsers break silently.
- **Absent is not zero**. A multiplier that does not apply should be absent from the payload, never present as `0` or `1`. The distinction between "no bonus" and "not applicable" matters to consumers.
- **Available on non-event days too**, returning `{"event": null}`. Silence is ambiguous with failure.

```
{
  "serverDay": "2026-12-18",
  "event": {
    "id": "evt-2026-12-18",
    "name": "COMBAT EVENT - Marked Reaver's Hour",
    "theme": "COMBAT", "rank": 2, "rankName": "Rising", "intensity": 6,
    "signature": "A4II-B19I-C7I-D15II",
    "modules": [
      { "slot": "A", "id": "A4", "magnitude": 2,
        "effects": { "combatXpMultiplier": 1.70 } },
      { "slot": "B", "id": "B19", "magnitude": 1, "target": "Aberrant",
        "effects": { "markedFamilyMultiplier": 3.0, "otherFamilyMultiplier": 0.6 } },
      { "slot": "C", "id": "C7", "magnitude": 1,
        "effects": { "setBonusPhantomPieces": 1 } },
      { "slot": "D", "id": "D15", "magnitude": 2,
        "effects": { "lootEchoChance": 0.12 } }
    ],
    "startsAt": "2026-12-18T17:00:00Z", "endsAt": "2026-12-19T17:00:00Z"
  }
}
```

Every consumer — the in-game banner, the event archive, the wiki, and the community calculators — reads this one payload. Nothing re-derives the event from the seed in production, even though it could; the seed is for verification and for showing players *why* today is what it is, not for parallel implementations that can drift.

19 Operations, testing, rollout

What to test

CHECK	ASSERTION
Determinism	The same seed and history produce a byte-identical payload on every shard and in the client port. Assert against stored fixtures.
PRNG agreement	Server and client agree to 12 decimal places over at least 6 draws from at least 4 seeds. This catches the int32 truncation bug described in §17.
Gate coherence	Every Focus in a filtered pool overlaps both the Boon and the Twist, and every excluded one fails a rule. Then disable each rule and confirm the assertions fail.
Non-empty pool	For all 400 Boon × Twist pairs, the filtered Focus pool is non-empty. This is a finite exhaustive check — run it, do not reason about it.
Clamp ceiling	Over a long simulated run, no resolved quantity exceeds ×4 (×6 at Rank IV).
Distribution	Over 10,000 simulated events: every module and every theme appears, and rank shares are within tolerance of \$5.

A check that returns an empty list on success only means something if you can tell "empty" from "did not run" — make the runner report how many assertions it executed, not just how many failed.

Telemetry worth having from day one

- Focus pool size per event — the health metric for the gate. If the mean drifts below about 5, the tables have grown too narrow and events will feel samey.
- Per-module appearance counts. Any module at zero after a simulated decade is unreachable and is a bug in its surfaces or tags.
- Active players and session length on event days against the trailing non-event baseline, split by Rank — this tells you whether Rank is landing.
- Which Boon/Twist pairs produce the smallest pools — the candidates for a new Focus module.

Suggested rollout

1. **Shadow mode.** Run the roll daily, publish nothing, log everything, for two to four weeks. Check that the events that came up sound like days you would want to ship.
2. **Reduced pools.** Launch with a subset — say twelve modules per slot — chosen for effects that already exist in the game and need no new systems. The structure count is still over 1.6 million.
3. **Widen.** Append modules as their underlying features land. Because pools are additive and the gate is structural, that requires no other changes.

PLAYER-FACING COMMUNICATION

Publish the module list and the tag rules. The system is far more interesting to players who can see the machine — an archive page listing every past event with its signature, its rank and its numeral turns a random daily buff into something with history that players collect and compare. It also makes Echo of Days (D20) legible instead of confusing.

20 Open questions for the developer

These are the decisions this document deliberately does not make, because they depend on the live game and on your intent rather than on the structure.

1. **Additive or multiplicative with gear?** The single biggest balance question (§10). The $\times 4$ ceiling assumes additive.
2. **Are the ten skills the right parameter pool?** The skill list used here is illustrative and must be bound to real skill IDs. Likewise the 34 zones, 12 gear slots, 8 enemy families, 8 resource families and 8 damage types — several of these are inferred from the existing toolkit and need confirming against the live data model.
3. **Which modules need new systems?** Roughly a dozen — the server pot (B12), the wandering merchant (B15), the temporary rift (D14), the event shop (D5), leaderboards (B13), the server goal (D7) — are meaningful engineering in their own right. They are worth keeping in the tables as targets, but the reduced-pool launch in §19 exists so they are not blockers.
4. **Does the death penalty exist in the form A10 assumes?** Several Twists assume there is a cost to dying worth removing or raising. If there is not, B5, B11 and A10 need rewriting.
5. **Should events be per-shard or global?** This spec assumes global, which is why the seed is the server-day string alone. Per-shard events need the shard ID in the seed and change the social character of the system considerably.
6. **What happens to an event in progress across a patch?** If module effects are read live from config, a mid-day deploy changes the event under players. Freeze the payload at reset and read from the frozen copy.

WHAT IS ALREADY DONE

Eighty modules with effects, tags, surfaces and three magnitude bands each. The gate, the clamps, the naming system, the anti-repeat memories and the distribution tuning. A verified deterministic PRNG with a matching JS port. A 100-year simulation harness and the numbers it produces. What remains is binding the effects to real game systems and deciding the six questions above.